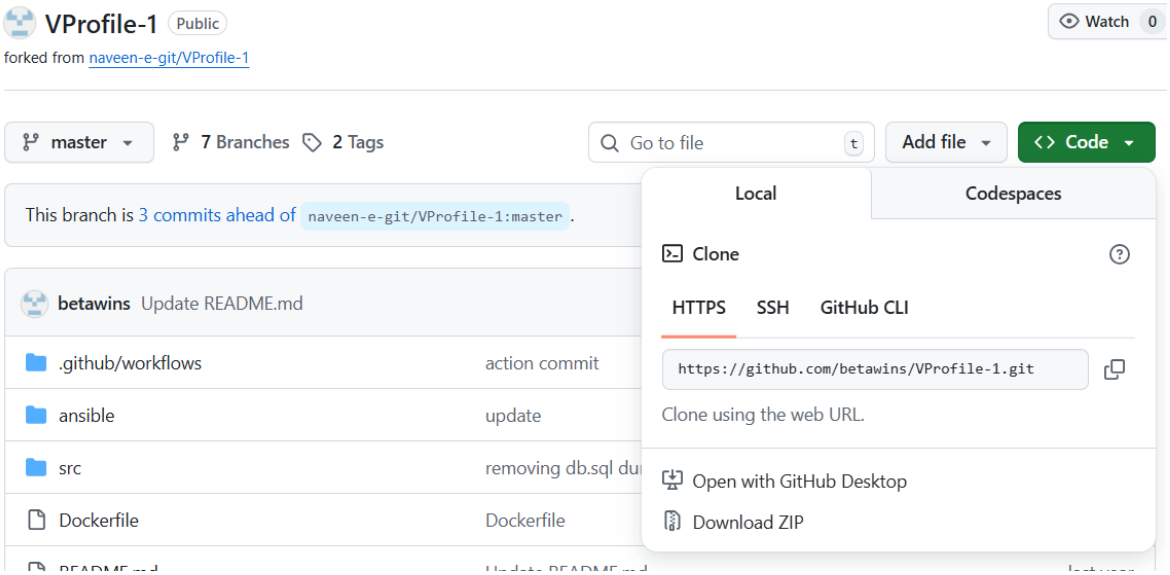
Task 1

Create one Jenkins job using the below code and create three stages:

- **stage1**: Git clone to download the source code.
- **stage2**: Sonarqube Integration to check the quality of code.
- **stage3**: Slack Integration to send the alerts to slack.

URL: https://github.com/betawins/VProfile-1.git

STAGE - 1

- open github
- find repo
- copy source code from github
- 

- Launch Jenkins instance and systemctl start jenkins
- Login to Jenkins
- New item > free style
- Source code paste ssh code in git column
- Apply and save it
- Build it

Console Output

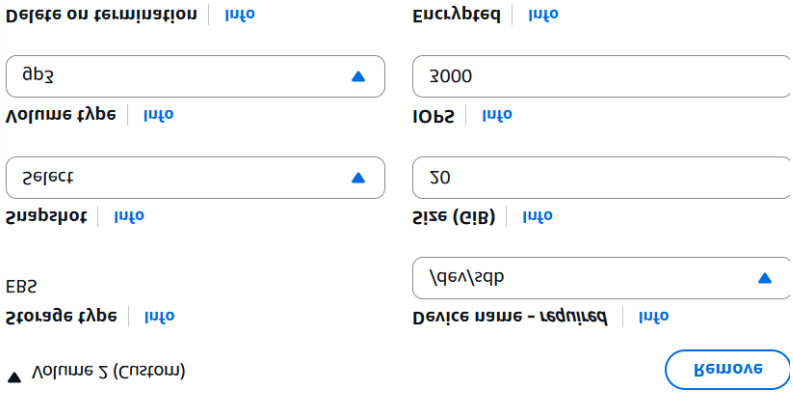
```
Started by user admin
Running as SYSTEM
Building on the built-in node in workspace /var/lib/jenkins/workspace/v-profile
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/VProfile-1.git
 > /usr/bin/git init /var/lib/jenkins/workspace/v-profile # timeout=10
Fetching upstream changes from https://github.com/betawins/VProfile-1.git
 > /usr/bin/git --version # timeout=10
 > git --version # 'git version 2.50.1'
 > /usr/bin/git fetch --tags --force --progress -- https://github.com/betawins/VProfile-1.git
 > /usr/bin/git config remote.origin.url https://github.com/betawins/VProfile-1.git # timeout=
 > /usr/bin/git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeou
Avoid second fetch
 > /usr/bin/git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision afc137167b41aaaffeaef82b81aee721ec883901 (refs/remotes/origin/master)
 > /usr/bin/git config core.sparsecheckout # timeout=10
 > /usr/bin/git checkout -f afc137167b41aaaffeaef82b81aee721ec883901 # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
Finished: SUCCESS
```

- Verify

```
[ec2-user@ip-10-0-1-110 v-profile]$ cd ..
[ec2-user@ip-10-0-1-110 workspace]$ ls
build   first-job  git-sync      jenkins      periodically  private-repo        repo-clone-private@tmp  v-profile
deploy  git-repo   git-sync@tmp  jenkins@tmp  poll-scm      repo-clone-private  test
[ec2-user@ip-10-0-1-110 workspace]$
```

STAGE -2 SONARQUBE

- Now open aws console
- Launch an instance
- Add instace type as t2.large
- Storage of 20gb



- Launch instance.
- Install java 17

```
Installed:
  java-17-amazon-corretto-devel-1:17.0.17+10-1.amzn2023.1.x86_64

Complete!
[ec2-user@ip-10-0-1-110 opt]$ java -version
openjdk version "17.0.17" 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-17.0.17.10.1 (build 17.0.17+10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.17.10.1 (build 17.0.17+10-LTS, mixed mode, sharing)
[ec2-user@ip-10-0-1-110 opt]$
```

- **i-0a4cbcc0493321880 (Jenkins)**
  PublicIPs: 54.160.238.95 PrivateIPs: 10.0.1.110
- Install docker
- Sudo yum install docker -y
- Now systemctl start docker
- Now, pull sonar
- Docker pull sonarqube:10.3-community

```
[root@ip-172-30-0-159 ec2-user]# docker pull sonarqube:10.3-community
10.3-community: Pulling from library/sonarqube
31bd5f451a84: Pull complete
26611c45681a: Pull complete
7657bba016af: Pull complete
e5c532b68318: Pull complete
65e11da75820: Pull complete
7f728301e94e: Pull complete
3ff564f2dfed: Pull complete
4f4fb700ef54: Pull complete
Digest: sha256:60f86482c5413c0f00ddc4daf14593dab1ae4fc708328bf1e9537b23faf
0cfe0
Status: Downloaded newer image for sonarqube:10.3-community
docker.io/library/sonarqube:10.3-community
```

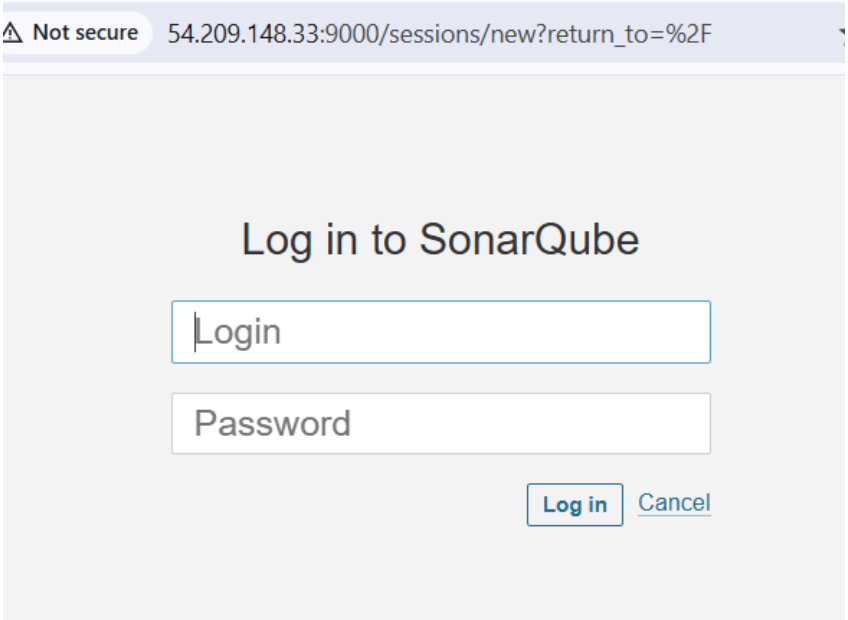
- To run docker
- Docker run -itd -p 9000:9000 sonarqube:10.3-community

```
[root@ip-172-30-0-159 ec2-user]# docker run -itd -p 9000:9000 sonarqube:10
3-community
79e16c78499fe0da0f96af50eb08f0a02ebdb9fbfe08a5845a220e5432b4cd08
[root@ip-172-30-0-159 ec2-user]# docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED
              STATUS              PORTS
    NAMES
79e16c78499f   sonarqube:10.3-community   "/opt/sonarqube/dock…"   About a
 minute ago   Up About a minute   0.0.0.0:9000->9000/tcp, :::9000->9000/tc
p   awesome_robinson
[root@ip-172-30-0-159 ec2-user]#
```

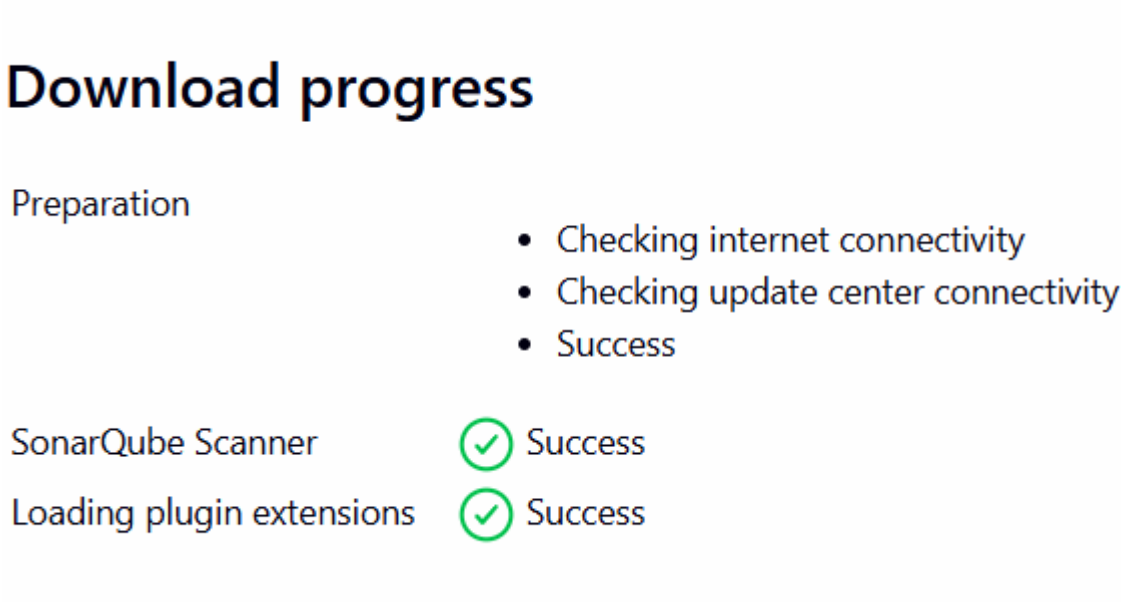
i-0b4c5b93ddec5b13a (sonarqube)

- Now allow 9000 port in inbound rules at instance level

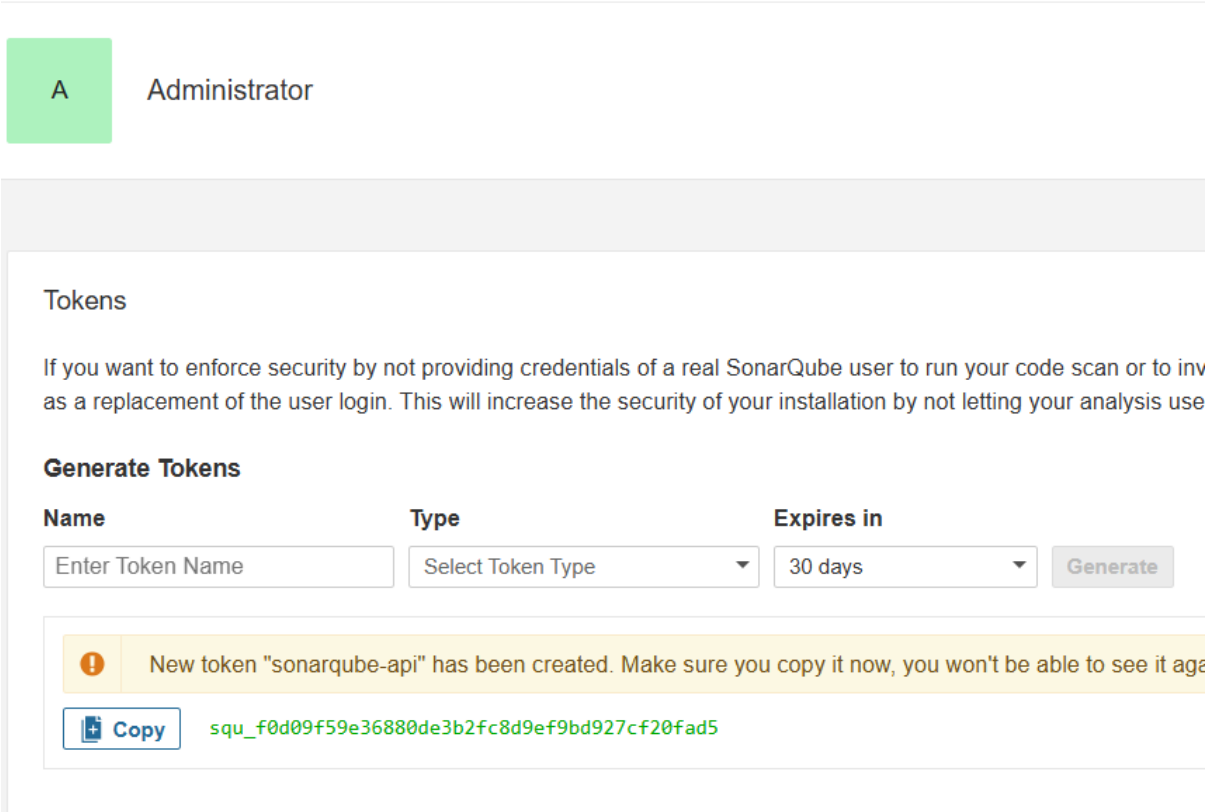
- And copy public ip of sonarqube server and add 9000 in web browser.
- 54.209.148.33:9000



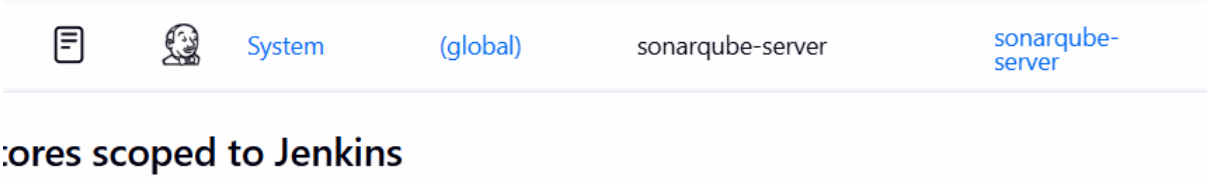
-
- Now login to Jenkins
- Go to manage Jenkins
- Add pulgins >> sonarqube scanner



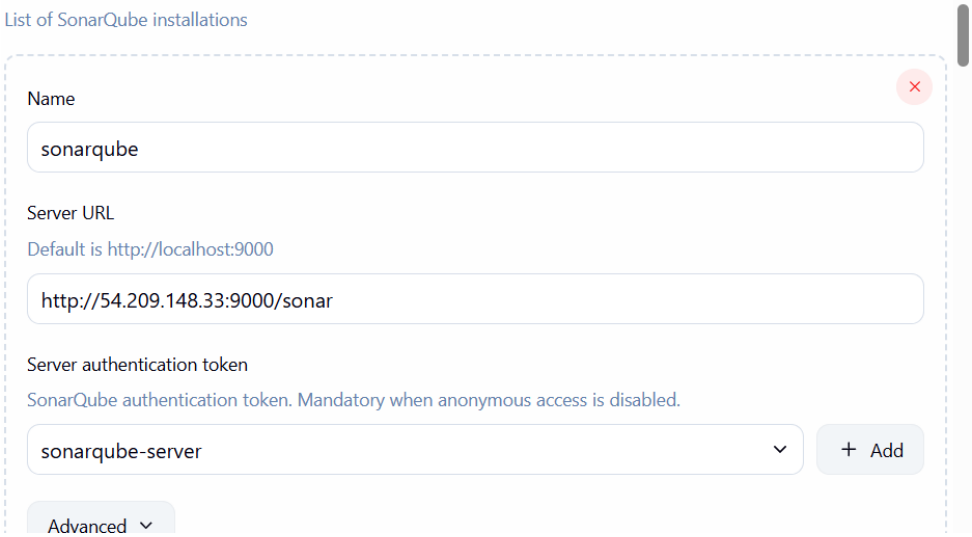
- Now go to sonarqube, my admin > security create token



- Copy token
- Navigate to Jenkins
- Go to manage Jenkins > then credentials
- Add sonarqube with secret key format
- Add token to it



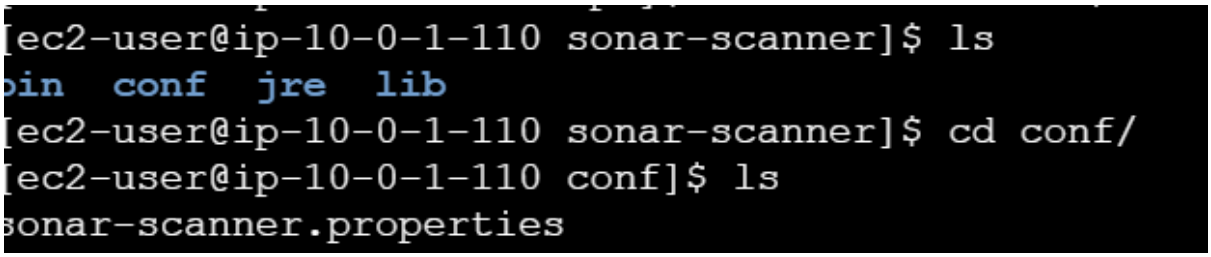
- Now go to manage Jenkins > system
- Search for sonarqube
- Add sonar url to it
- Atttach sonarqube credential dropdown

- 

- Now, install sonar scanner in Jenkins server
- cd /opt
- sudo wget https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-4.6.2.2472-linux.zip
- unzip it
- sudo unzip sonar-scanner-cli-4.6.2.2472-linux.zip
- change the name
- sudo mv sonar-scanner-4.6.2.2472-linux sonar-scanner

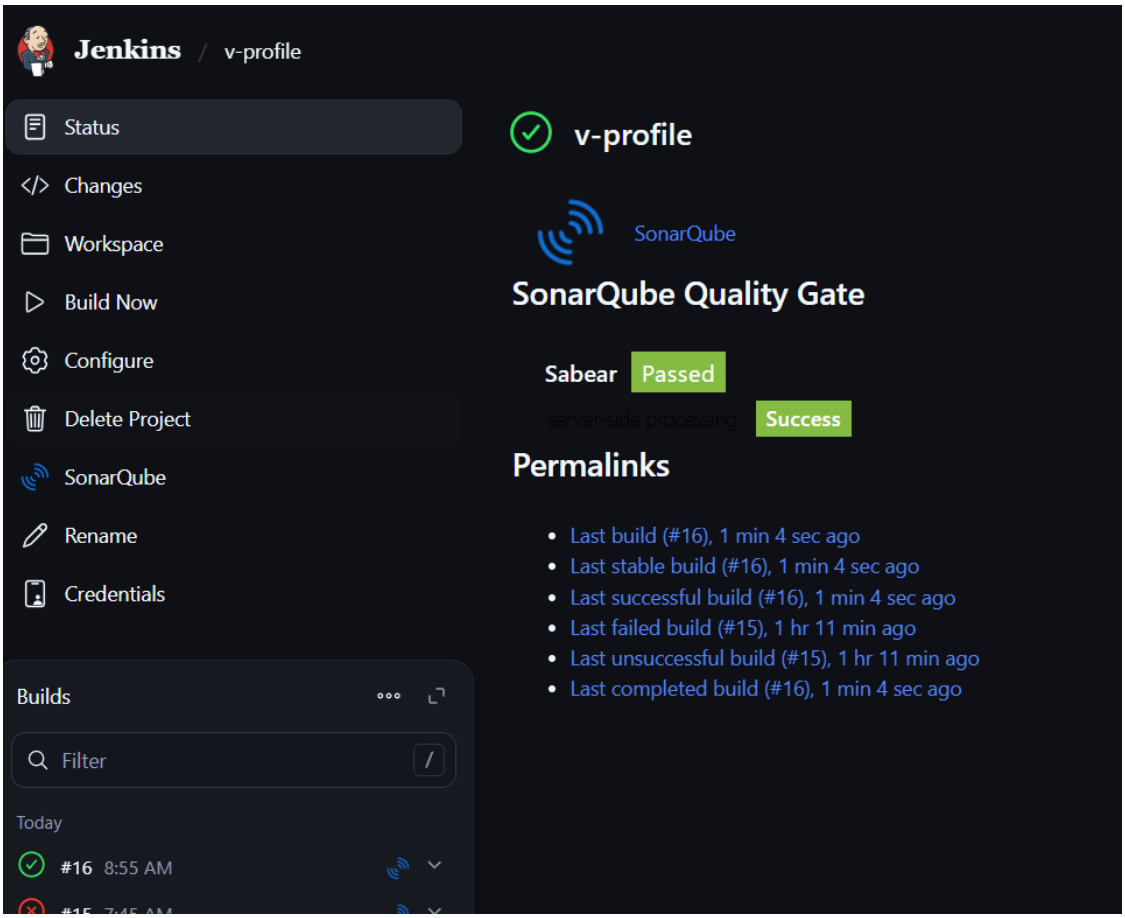
```
[ec2-user@ip-10-0-1-110 opt]$ ls
aws                 jenkins-backups   sonar-scanner-cli-4.6.2.2472-linux.zip
jenkins-backup.sh   sonar-scanner
[ec2-user@ip-10-0-1-110 opt]$
```

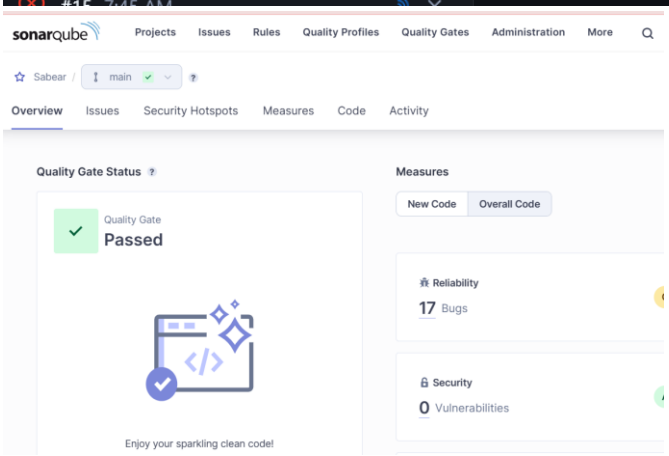
i-0a4cbcc0493321880 (Jenkins)

- now go to cd sonar-scanner
- ls
- 

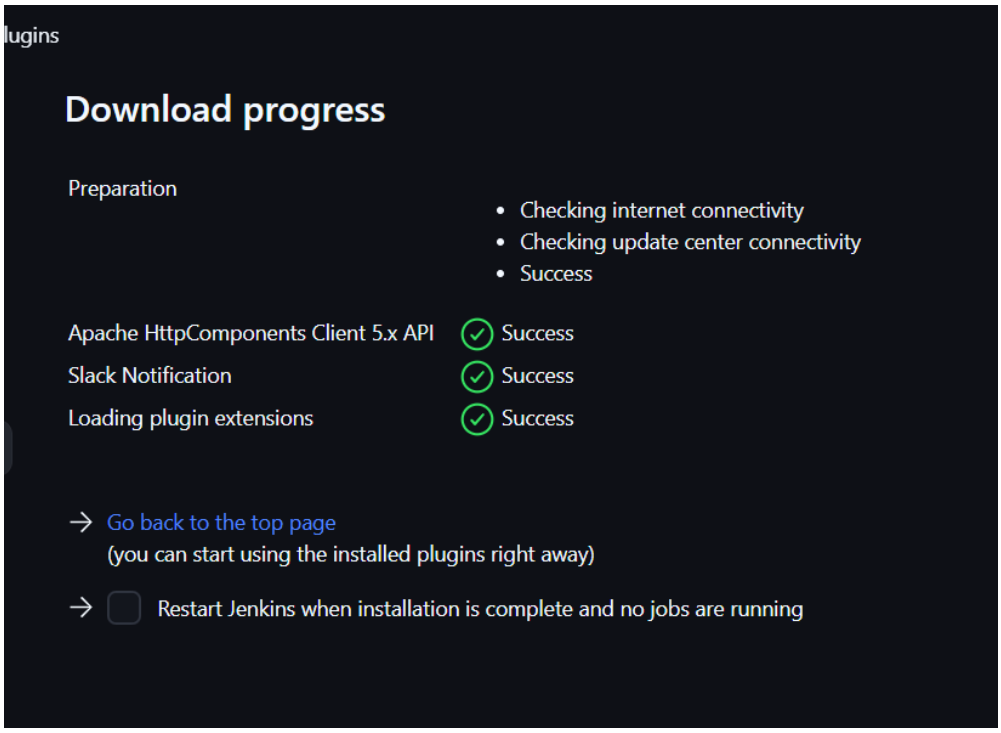
- Cd conf
- Ls
- Edit sonar-scanner.properties
- Add sonar url to it

```
#Configure here general information about the environment, such as SonarQub
e server connection details for example
#No information about specific project should appear here

#----- Default SonarQube server
#sonar.host.url=http://54.209.148.33:9000/sonar

#----- Default source code encoding
#sonar.sourceEncoding=UTF-8
```

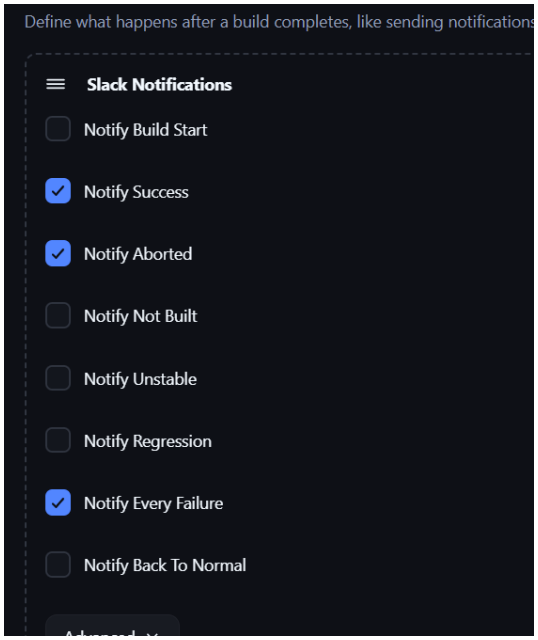
- Now copy the copy the location sonar scanner
- Add it in Jenkins > manage Jenkins > tools
- 

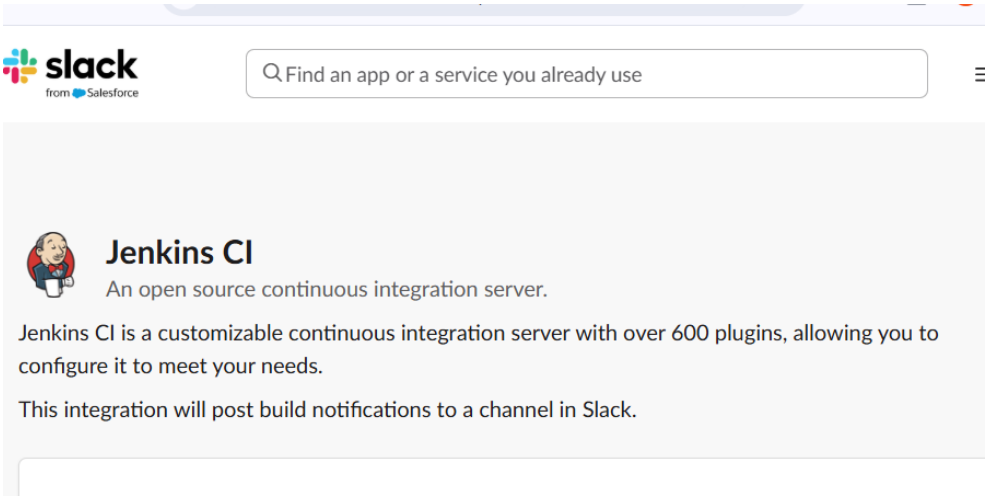
- 

- **stage3**: Slack Integration to send the alerts to slack
- add slack plugin
- in Jenkins
- Jenkins > manage Jenkins > pulgins
- >> slack notification

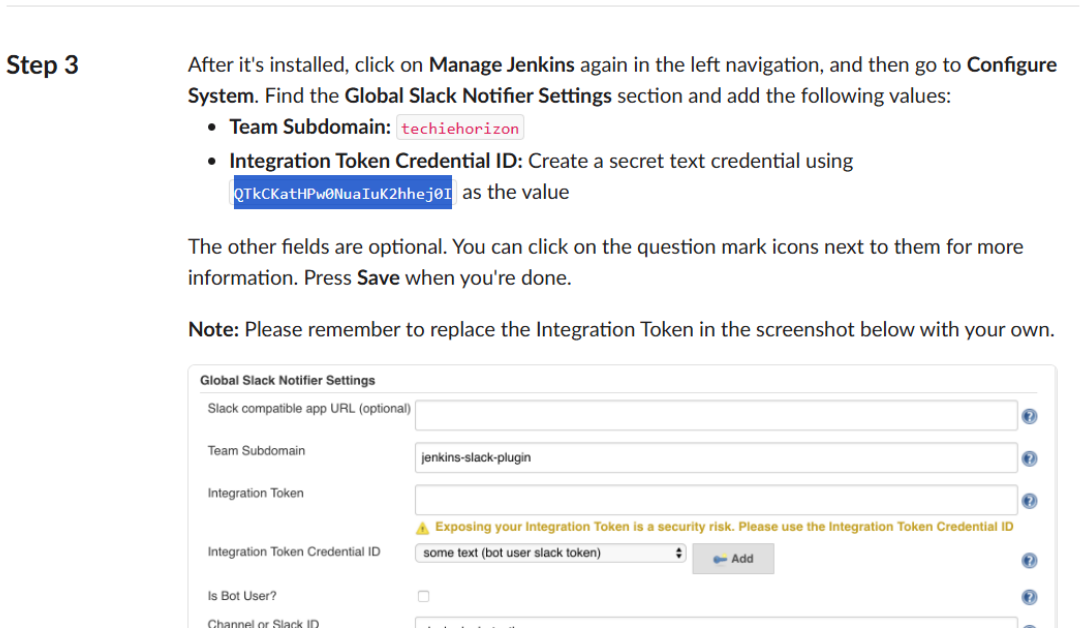


- Now > Jenkins > job > configure > build steps
- Select slack

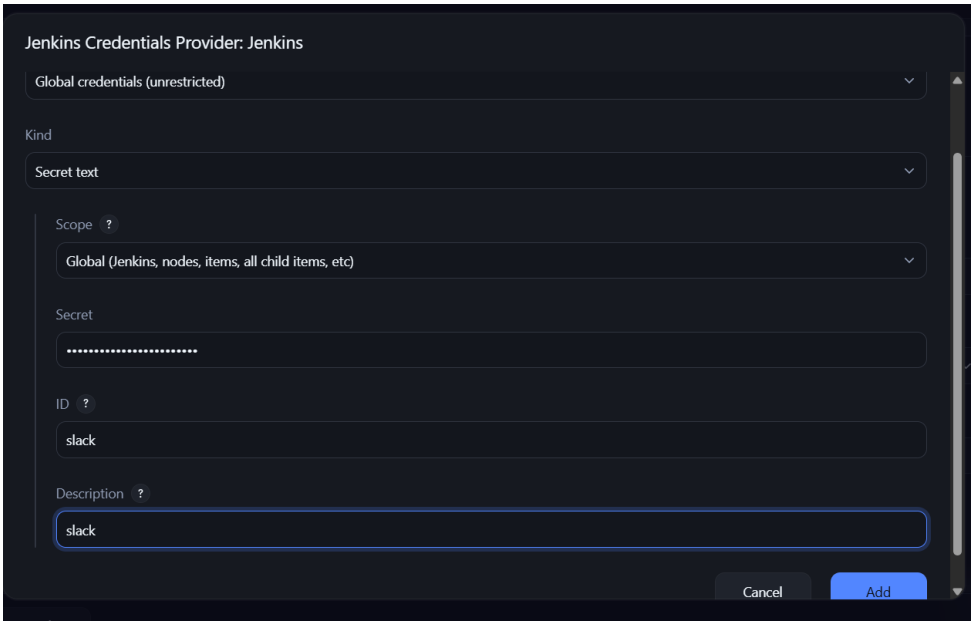


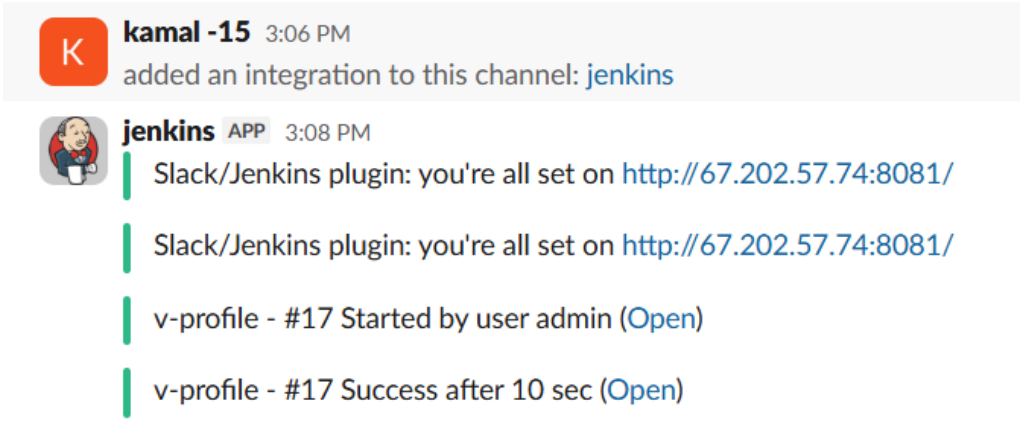
- 

- Add channel

- 

- Add subdomain name and credentials create in Jenkins with these

- 

- 

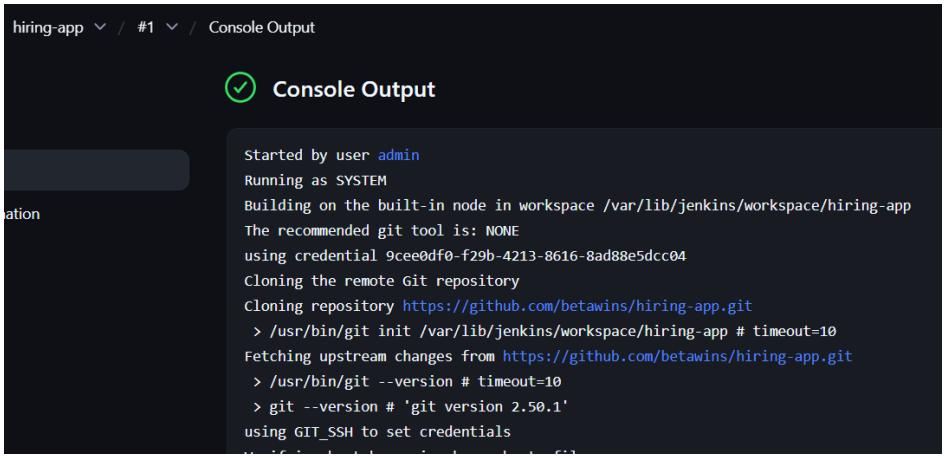
Task 2

Create one Jenkins job using the below code and create three stages:

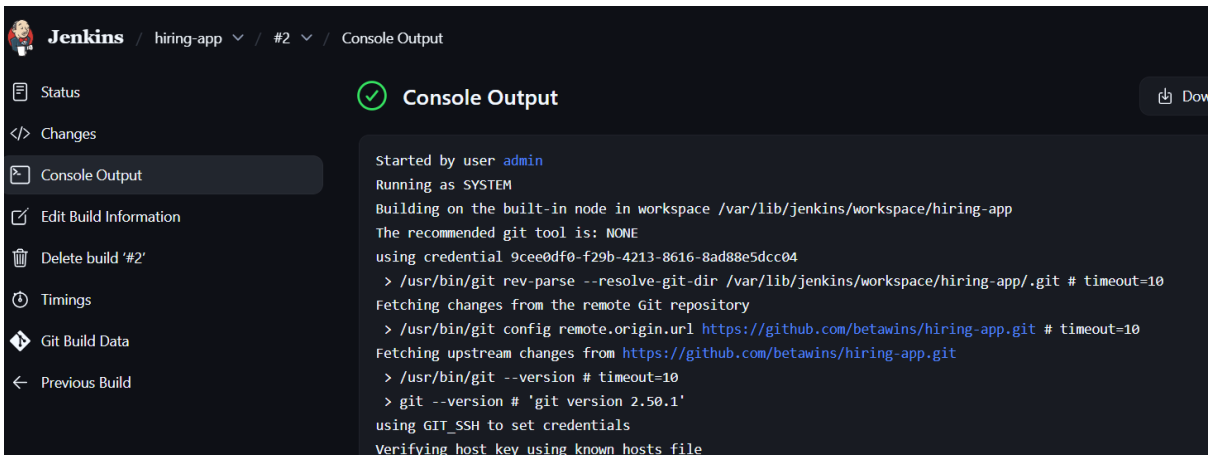
- **stage1**: Git clone to download the source code.
- **stage2**: Sonarqube Integration to check the quality of code.
- **stage3**: Slack Integration to send the alerts to slack.

URL: https://github.com/betawins/hiring-app.git

  - login to Jenkins
  - create a free style job
  - add git repo and add git branch

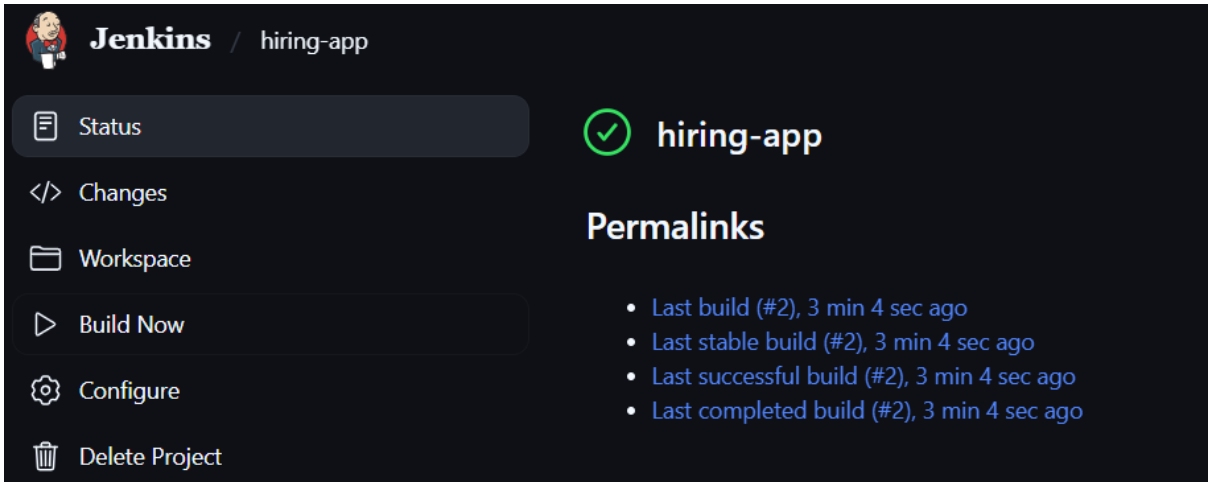


- now, go to sonarqube,
- admin
- security
- generate token
- now add token in Jenkins
- credentials
- add url , secret key type ,
- and add token
- ----------create credential
- Now, add this credentials to sonar server
- Build it



- Now slack integration to hiring app
- Go to job >
- Configure
- Scroll down to build steps
- Add slack notification

- Build it


Jenkins
hiring-app
Status
Changes
Workspace
Build Now
Configure
Delete Project
hiring-app
Permalinks
Last build (#2), 3 min 4 sec ago
Last stable build (#2), 3 min 4 sec ago
Last successful build (#2), 3 min 4 sec ago
Last completed build (#2), 3 min 4 sec ago